MODULE 3 Rational Numbers

Key Vocabulary

rational number *(número racional)*

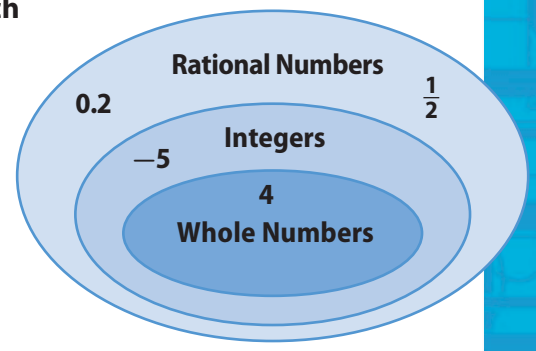
Venn diagram *(diagrama de Venn)*

ESSENTIAL QUESTION

How can you use rational numbers to solve real-world problems?

EXAMPLE 1

Use the Venn diagram to determine in which set or sets each number belongs.



A. $\frac{1}{2}$ belongs in the set of rational numbers.

B. −5 belongs in the sets of integers and rational numbers.

C. 4 belongs in the sets of whole numbers, integers, and rational numbers.

D. 0.2 belongs in the set of rational numbers.

EXAMPLE 2

Order $\frac{2}{5}$, 0.2, and $\frac{4}{15}$ from greatest to least.

Write the decimal as an equivalent fraction. $0.2 = \frac{2}{10} = \frac{1}{5}$

Find equivalent fractions with 15 as the common denominator. $\frac{2 \times 3}{5 \times 3} = \frac{6}{15}$ $\quad \frac{1 \times 3}{5 \times 3} = \frac{3}{15}$ $\quad \frac{4}{15} = \frac{4}{15}$

Order fractions with common denominators by comparing the numerators. $6 > 4 > 3$ $\quad \frac{6}{15} > \frac{4}{15} > \frac{3}{15}$

The numbers in order from greatest to least are, $\frac{2}{5}$, $\frac{4}{15}$, and 0.2.

EXERCISES

Classify each number by indicating in which set or sets it belongs. (Lesson 2.1)

1. 8 _____

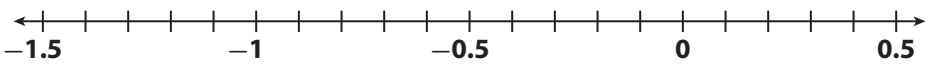
2. 0.25 _____

Find the absolute value of each rational number. (Lesson 2.2)

3. $|3.7|$ _____ **4.** $\left|-\frac{2}{3}\right|$ _____

Graph each set of numbers on the number line and order the numbers from greatest to least. (Lessons 2.1, 2.3)

5. $-0.5, -1, -\frac{1}{4}, 0$



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